Advanced Network Security and Architectures

METI

Project Report [EN]

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Contents

1 Introduction

Group Encrypted Transport VPN (GETVPN) is a Cisco proprietary technology designed to provide scalable, any-to-any encrypted communication over private WAN infrastructures such as MPLS networks. Unlike traditional point-to-point IPSec VPNs, which require a dedicated tunnel between each pair of sites, GETVPN operates on a group key model: all participating routers share the same encryption keys and policy, allowing any member to securely communicate with any other member without the overhead of per-pair tunnel establishment.

The GETVPN architecture is built around two roles. The **Key Server (KS)** is responsible for generating and distributing cryptographic keys and security policies to all group participants. The **Group Members (GMs)** are the routers that register with the Key Server, receive the shared keys and traffic policies, and then encrypt or decrypt data traffic autonomously. Once registered, GMs communicate directly with each other; the Key Server is not involved in the data plane.

The goal of this project is to deploy and study a GETVPN solution in a simulated environment using GNS3, analyse the key management protocol exchanges, observe traffic encryption behaviour, and extend the base configuration with additional features. The remainder of this report describes the topology, the configuration steps, the observations made during the lab tasks, and the extensions implemented.

2 Topology and Base Configuration

The network topology used in this project is shown in Figure 1. It consists of four Cisco IOSv routers interconnected through a central switch. Router R1 acts as the Key Server and is connected to the shared segment `192.168.1.0/24` with address `192.168.1.254`. Routers R2, R3, and R4 are the Group Members, with addresses `192.168.1.2`, `192.168.1.3`, and `192.168.1.4` respectively on their `gi0/0` interfaces. Each Group Member also has a LAN-facing interface (`gi0/1`) connecting to a dedicated subnet: R2 serves `192.168.10.0/24`, R3 serves `192.168.20.0/24`, and R4 serves `192.168.30.0/24`.

Three end hosts (PC1, PC2, PC3) are attached to these LAN subnets. Since the lab environment does not include dedicated PC images, Cisco 3725 routers were used to simulate the hosts; IP routing was disabled on these devices (`no ip routing`), and a default gateway static route was configured on each to point to the respective Group Member. Each simulated PC was assigned the `.100` address of its subnet (e.g., PC1 at `192.168.10.100`) and its icon was changed to a PC in GNS3 for clarity.

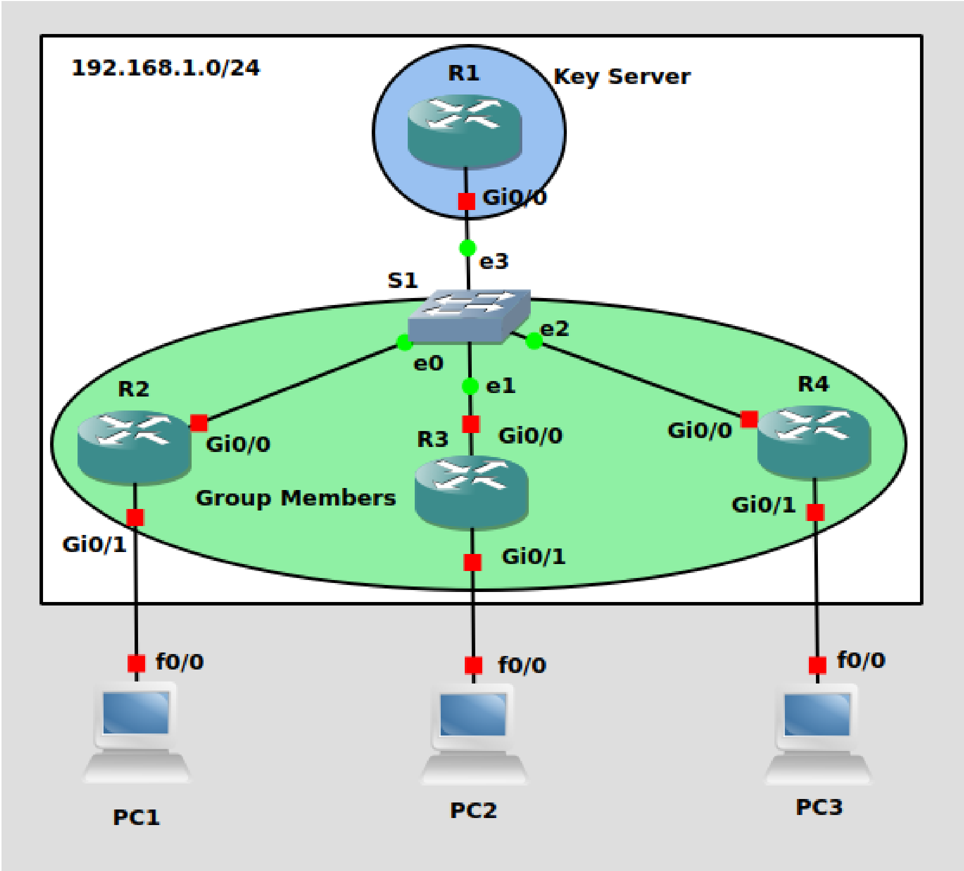


Figure 1: Project topology. R1 is the Key Server; R2, R3, and R4 are Group Members.

OSPF was chosen as the routing protocol and was configured on all four routers to advertise all connected networks. Prior to applying any GETVPN or IPSec configuration, end-to-end reachability was verified by pinging between all three PCs and between all router interfaces. All pings were successful, confirming that the routing layer was fully operational before any security configuration was introduced.

3 GETVPN Configuration

3.1 Overview of the Configuration Model

GETVPN security configuration is divided between the Key Server and the Group Members. The Key Server defines all security policy — the IKE parameters, the cryptographic algorithms, the traffic selector (which packets to encrypt), and the keys themselves — and distributes this policy to GMs during their registration. GMs only need to know how to authenticate with the KS and where to register; they receive everything else from the KS automatically.

The key management protocol used to distribute keys and policy is **GDOI (Group Domain of Interpretation)**, which runs over UDP port 848. The registration process has two phases: an IKE Phase 1 exchange that establishes a secure authenticated channel between the GM and the KS, followed by a **GROUPKEY-PULL** exchange in which the GM requests and receives the group keys and IPSec policy.

3.2 Key Server Configuration (R1)

3.2.1 IKE Phase 1 Policy

The first step is to define the parameters for the IKE Phase 1 exchange that will protect the GDOI registration channel between each GM and the KS. AES encryption, SHA hashing, pre-shared key authentication, and Diffie-Hellman group 5 are used. The wildcard address `0.0.0.0` allows the KS to accept registrations from any GM without listing each one individually.

```
! IKE Phase 1
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes
R1(config-isakmp)#hash sha
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config)#crypto isakmp key saar address 0.0.0.0
```

3.2.2 Transform Set and IPSec Profile

The *transform set* defines the cryptographic algorithms that will be applied to data traffic: ESP with AES for encryption and SHA-HMAC for integrity. The *IPSec profile* bundles the transform set into a named object that is later referenced inside the GDOI group configuration.

```
! Transform set (cipher suite for data traffic)
R1(config)#crypto ipsec transform-set myTSet esp-aes esp-sha-hmac

! IPSec profile (references the transform set)
R1(config)#crypto ipsec profile myIPSecProfile
R1(ipsec-profile)#set transform-set myTSet
```

3.2.3 RSA Key Pair

The KS signs its rekey messages using an RSA key pair so that GMs can verify the authenticity and integrity of key updates. The label `myRSAKeys` is used to reference this key pair in the GDOI group configuration.

```
R1(config)#crypto key generate rsa modulus 1024 label myRSAKeys
```

3.2.4 Traffic Selector ACL

GETVPN does not encrypt all traffic indiscriminately. An extended ACL defines the *interesting traffic* — the flows that should be protected. This ACL is configured on the KS and pushed to all GMs during registration. In this lab, only ICMP traffic is protected, which is sufficient for the ping-based verification tasks.

```
! Interesting traffic (what gets encrypted)
R1(config)#ip access-list extended ICMP
R1(config-ext-nacl)#permit icmp any any
```

3.2.5 GDOI Group

The GDOI group is the central object of the KS configuration. It ties together the identity number (which GMs must match to join the group), the KS address, the rekey mechanism, and the IPSec SA policy including the traffic selector.

```
R1(config)#crypto gdoi group myGDOIGroup
R1(config-gkm-group)#identity number 123
R1(config-gkm-group)#server local
R1(gkm-local-server)#address ipv4 192.168.1.254
R1(gkm-local-server)#rekey authentication mypubkey rsa myRSAKeys
R1(gkm-local-server)#rekey transport unicast
R1(gkm-local-server)#sa ipsec 10
R1(gkm-sa-ipsec)#profile myIPSecProfile
R1(gkm-sa-ipsec)#match address ipv4 ICMP
```

The `rekey transport unicast` directive instructs the KS to send key refresh messages directly to each registered GM individually, as opposed to multicast delivery.

3.3 Group Member Configuration (R2, R3, R4)

The configuration is identical on all three GMs; only the hostname differs. Each GM configures the same IKE Phase 1 parameters and pre-shared key as the KS, joins the GDOI group by its identity number and the KS address, and applies a crypto map to its WAN-facing interface to activate the received policy.

```
! IKE Phase 1 (must match KS policy)
Rx(config)#crypto isakmp policy 10
Rx(config-isakmp)#encryption aes
Rx(config-isakmp)#hash sha
Rx(config-isakmp)#authentication pre-share
Rx(config-isakmp)#group 5
Rx(config)#crypto isakmp key saar address 192.168.1.254

! Transform set (required locally, actual policy comes from KS)
Rx(config)#crypto ipsec transform-set TRANSFORM_SET esp-aes esp-sha-hmac

! GDOI group (must match KS identity number)
Rx(config)#crypto gdoi group myGDOIGroup
Rx(config-gkm-group)#identity number 123
Rx(config-gkm-group)#server address ipv4 192.168.1.254

! Crypto map applied to WAN interface
Rx(config)#crypto map myCryptoMap 10 gdoi
Rx(config-crypto-map)#set group myGDOIGroup
Rx(config)#interface gi0/0
Rx(config-if)#crypto map myCryptoMap
```

It is worth noting that the transform set defined locally on a GM is not used for data traffic; the actual IPSec policy (including transform set, traffic selector, and keys) is always received from the KS. The local transform set is only relevant for the IKE Phase 1 exchange during registration.

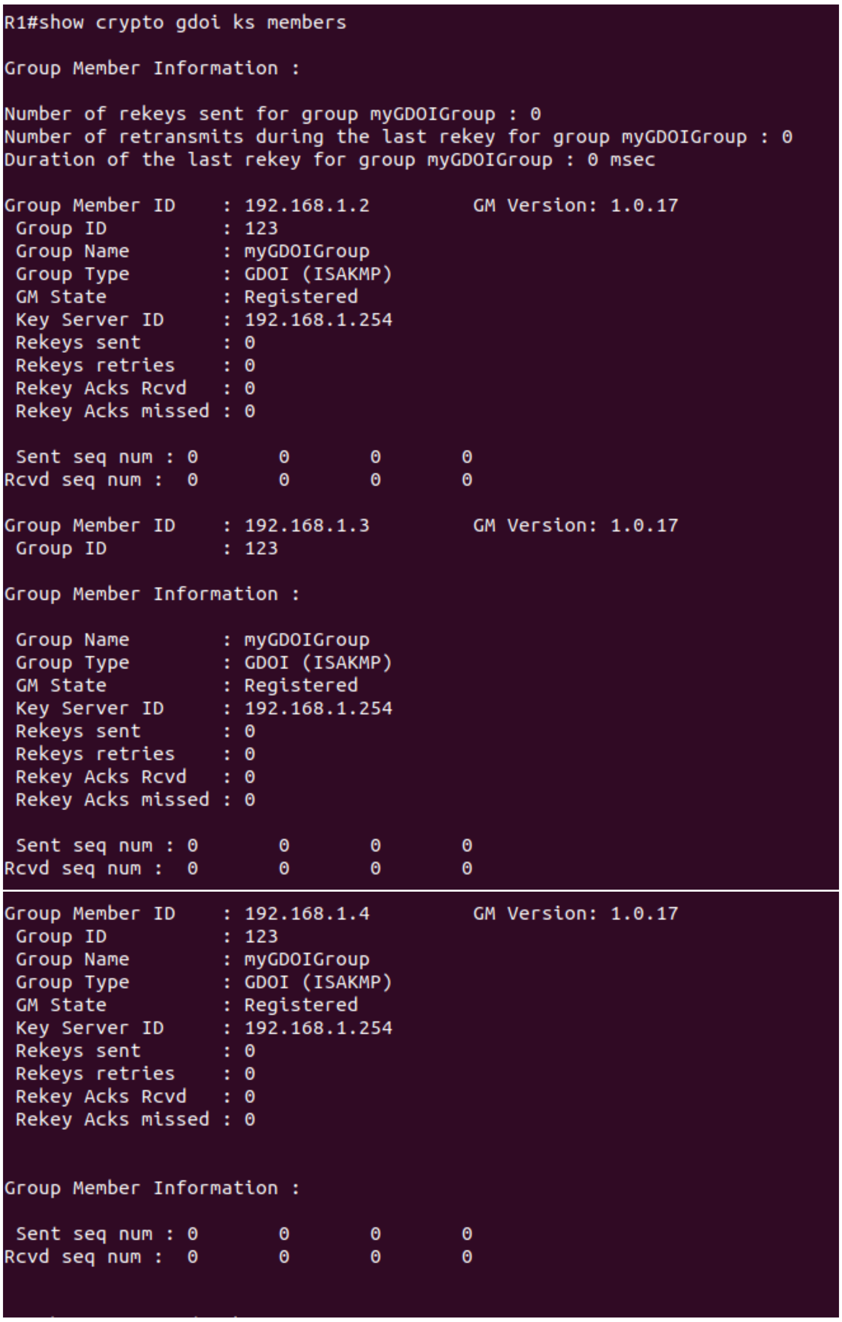
4 Verification

4.1 Key Server Status

After all GMs were configured, the `show crypto gdoi` and `show crypto gdoi ks members` commands were used on R1 to confirm correct operation.

```
R1#show crypto gdoi
GROUP INFORMATION

    Group Name                 : myGDOIGroup (Unicast)
    Re-auth on new CRL         : Disabled
    Group Identity             : 123
    Group Type                 : GDOI (ISAKMP)
    Crypto Path                : ipv4
    Key Management Path        : ipv4
    Group Members              : 3
    IPSec SA Direction         : Both
    IP D3P Window              : Disabled
    CKM status                 : Disabled
    Group Rekey Lifetime       : 86400 secs
    Group Rekey
        Remaining Lifetime     : 86355 secs
        Time to Rekey          : 86130 secs
        Acknowledgement Cfg    : Cisco
    Rekey Retransmit Period    : 10 secs
    Rekey Retransmit Attempts: 2
    Group Retransmit
        Remaining Lifetime     : 0 secs

      IPSec SA Number          : 10
      IPSec SA Rekey Lifetime: 3600 secs
      Profile Name             : myIPSecProfile
      Replay method            : Count Based
      Replay Window Size       : 64
      Tagging method           : Disabled
      SA Rekey
         Remaining Lifetime    : 3556 secs
        Time to Rekey          : 3170 secs
      ACL Configured           : access-list ICMP

     Group Server list         : Local
```

Figure 2: `show crypto gdoi` on the Key Server (R1). Three Group Members registered.



```
R1#show crypto gdoi ks members

Group Member Information :

Number of rekeys sent for group myGDOIGroup : 0
Number of retransmits during the last rekey for group myGDOIGroup : 0
Duration of the last rekey for group myGDOIGroup : 0 msec

Group Member ID    : 192.168.1.2        GM Version: 1.0.17
 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0

Group Member ID    : 192.168.1.3        GM Version: 1.0.17
 Group ID          : 123

Group Member Information :

 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0

Group Member ID    : 192.168.1.4        GM Version: 1.0.17
 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0


Group Member Information :

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0
```

Figure 3: `show crypto gdoi ks members` on R1. All three GMs (`192.168.1.2`, `.3`, `.4`) show `GM State: Registered`.

The output confirms that all three GMs successfully completed the GROUPKEY-PULL exchange. Each entry shows `GM State: Registered` and identifies the correct Key Server address (`192.168.1.254`). The rekey counters are all zero, which is expected since no TEK refresh has occurred yet.

4.2 Group Member Status

The `show crypto gdoi` and `show crypto gdoi ipsec sa` commands were run on R2 as a representative Group Member.

Several important aspects of GETVPN operation are visible in the output from Figure 4:

- **Registration status**: R2 shows `Registration status: Registered`, confirming it successfully joined the group.
- **ACL received from KS**: The traffic selector (`permit icmp any any`) was pushed by the KS and installed on R2 as `gdoi_group_myGDOIGroup_temp_acl`. GMs do not define this locally.
- **TEK policy**: The Traffic Encryption Key policy shows `transform:esp-aes esp-sha-hmac` and SPI `0x6D2F03B9`, with a remaining lifetime of 3478 seconds. This is the key used to encrypt ICMP traffic between GMs.
- **KEK policy**: The Key Encryption Key uses 3DES with a lifetime of 86276 seconds and is used to protect future rekey messages from the KS. Notably, the KEK uses 3DES regardless of the AES algorithm specified in the IKE policy; this is a known behaviour in the IOS version used.
- **Active TEK Number: 1**: Confirms one TEK is active and ready to encrypt matching traffic.

```
R2#show crypto gdoi
GROUP INFORMATION

    Group Name               : myGDOIGroup
    Group Identity           : 123
    Group Type               : GDOI (ISAKMP)
    Crypto Path              : ipv4
    Key Management Path      : ipv4
    Rekeys received          : 0
    IPSec SA Direction       : Both

     Group Server list       : 192.168.1.254

Group Member Information For Group myGDOIGroup:
    IPSec SA Direction       : Both
    ACL Received From KS     : gdoi_group_myGDOIGroup_temp_acl

    Group member             : 192.168.1.2      vrf: None
       Local addr/port       : 192.168.1.2/848
       Remote addr/port      : 192.168.1.254/848
       fvrf/ivrf             : None/None
       Version               : 1.0.17
       Registration status   : Registered
       Registered with       : 192.168.1.254
       Re-registers in       : 3229 sec
       Succeeded registration: 1
       Attempted registration: 1
       Last rekey from       : 0.0.0.0
       Last rekey seq num    : 0
       Unicast rekey received: 0
       Rekey ACKs sent       : 0
       Rekey Received        : never
       DP Error Monitoring   : OFF
       IPSEC init reg executed    : 0
       IPSEC init reg postponed   : 0
       Active TEK Number     : 1
       SA Track (OID/status) : disabled

    Rekeys cumulative
       Total received        : 0
       After latest register : 0
       Rekey Acks sents      : 0

 ACL Downloaded From KS 192.168.1.254:
   access-list   permit icmp any any

KEK POLICY:
    Rekey Transport Type     : Unicast
    Lifetime (secs)          : 86276
    Encrypt Algorithm        : 3DES
    Key Size                 : 192
    Sig Hash Algorithm       : HMAC_AUTH_SHA
    Sig Key Length (bits)    : 1296

TEK POLICY for the current KS-Policy ACEs Downloaded:
  GigabitEthernet0/0:
    IPsec SA:
        spi: 0x6D2F03B9(1831797689)
        KGS: Disabled
        transform: esp-aes esp-sha-hmac
        sa timing:remaining key lifetime (sec): (3478)
        Anti-Replay(Counter Based) : 64
        tag method : disabled
        alg key size: 16 (bytes)
        sig key size: 20 (bytes)
        encaps: ENCAPS_TUNNEL
```

Figure 4: `show crypto gdoi` on R2. Registration confirmed; TEK and KEK policies downloaded from KS.

5 Lab Tasks

5.1 IKE and GROUPKEY-PULL Analysis

To capture the GDOI registration exchanges, a Wireshark probe was placed on the link between the Key Server (R1) and the central switch before any Group Member was started. R1 was brought up first, followed by R2, R3, and R4 in sequence. The traffic on this link runs on UDP port 848, which Wireshark does not decode automatically. Figure 5 shows the raw capture before applying any decoder, with packets appearing as generic UDP. The *Decode As* feature was used to instruct Wireshark to interpret port 848 traffic as ISAKMP, after which the exchange types became readable.

udp.port == 848

No.	Time	Source	Destination	Protocol	Length	Info
82	11.368100	192.168.1.2	192.168.1.254	UDP	210	848 → 848 Len=168
84	11.400215	192.168.1.254	192.168.1.2	UDP	150	848 → 848 Len=108
85	11.579027	192.168.1.2	192.168.1.254	UDP	414	848 → 848 Len=372
86	11.625902	192.168.1.254	192.168.1.2	UDP	410	848 → 848 Len=368
87	11.684127	192.168.1.3	192.168.1.254	UDP	210	848 → 848 Len=168
88	11.796866	192.168.1.2	192.168.1.254	UDP	150	848 → 848 Len=108
89	11.835710	192.168.1.254	192.168.1.2	UDP	118	848 → 848 Len=76
90	11.836983	192.168.1.254	192.168.1.3	UDP	150	848 → 848 Len=108
91	11.890377	192.168.1.3	192.168.1.254	UDP	414	848 → 848 Len=372
92	11.960476	192.168.1.4	192.168.1.254	UDP	210	848 → 848 Len=168
93	12.002840	192.168.1.2	192.168.1.254	UDP	134	848 → 848 Len=92
94	12.057683	192.168.1.254	192.168.1.3	UDP	410	848 → 848 Len=368
95	12.059283	192.168.1.254	192.168.1.4	UDP	150	848 → 848 Len=108
96	12.060734	192.168.1.254	192.168.1.2	UDP	278	848 → 848 Len=236
97	12.104609	192.168.1.3	192.168.1.254	UDP	150	848 → 848 Len=108
98	12.165482	192.168.1.4	192.168.1.254	UDP	414	848 → 848 Len=372
99	12.206568	192.168.1.2	192.168.1.254	UDP	102	848 → 848 Len=60
100	12.281874	192.168.1.254	192.168.1.3	UDP	118	848 → 848 Len=76
101	12.283891	192.168.1.254	192.168.1.2	UDP	390	848 → 848 Len=348
102	12.285297	192.168.1.254	192.168.1.4	UDP	410	848 → 848 Len=368
103	12.310353	192.168.1.3	192.168.1.254	UDP	134	848 → 848 Len=92
104	12.387496	192.168.1.4	192.168.1.254	UDP	150	848 → 848 Len=108
105	12.493902	192.168.1.254	192.168.1.4	UDP	118	848 → 848 Len=76
106	12.495173	192.168.1.254	192.168.1.3	UDP	278	848 → 848 Len=236
107	12.519393	192.168.1.3	192.168.1.254	UDP	102	848 → 848 Len=60
108	12.593413	192.168.1.4	192.168.1.254	UDP	134	848 → 848 Len=92
109	12.702747	192.168.1.254	192.168.1.3	UDP	390	848 → 848 Len=348
110	12.704298	192.168.1.254	192.168.1.4	UDP	278	848 → 848 Len=236
111	12.796863	192.168.1.4	192.168.1.254	UDP	102	848 → 848 Len=60
112	12.915493	192.168.1.254	192.168.1.4	UDP	390	848 → 848 Len=348

Figure 5: Capture filtered on UDP port 848 before applying the ISAKMP decoder.

udp.port == 848

No.	Time	Source	Destination	Protocol	Length	Info
42	40.060597	192.168.1.2	192.168.1.254	ISAKMP	210	Identity Protection (Main Mode)
43	40.062094	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
44	40.063573	192.168.1.2	192.168.1.254	ISAKMP	414	Identity Protection (Main Mode)
45	40.068368	192.168.1.254	192.168.1.2	ISAKMP	410	Identity Protection (Main Mode)
46	40.073196	192.168.1.2	192.168.1.254	ISAKMP	150	Identity Protection (Main Mode)
47	40.074337	192.168.1.254	192.168.1.2	ISAKMP	118	Identity Protection (Main Mode)
48	40.075796	192.168.1.2	192.168.1.254	ISAKMP	134	Quick Mode
49	40.077498	192.168.1.254	192.168.1.2	ISAKMP	278	Quick Mode
50	40.078929	192.168.1.2	192.168.1.254	ISAKMP	102	Quick Mode
51	40.080272	192.168.1.254	192.168.1.2	ISAKMP	390	Quick Mode
81	57.704422	192.168.1.3	192.168.1.254	ISAKMP	210	Identity Protection (Main Mode)
82	57.705650	192.168.1.254	192.168.1.3	ISAKMP	150	Identity Protection (Main Mode)
83	57.707112	192.168.1.3	192.168.1.254	ISAKMP	414	Identity Protection (Main Mode)
84	57.711980	192.168.1.254	192.168.1.3	ISAKMP	410	Identity Protection (Main Mode)
85	57.717009	192.168.1.3	192.168.1.254	ISAKMP	150	Identity Protection (Main Mode)
86	57.718237	192.168.1.254	192.168.1.3	ISAKMP	118	Identity Protection (Main Mode)
87	57.719704	192.168.1.3	192.168.1.254	ISAKMP	134	Quick Mode
88	57.721103	192.168.1.254	192.168.1.3	ISAKMP	278	Quick Mode
89	57.722382	192.168.1.3	192.168.1.254	ISAKMP	102	Quick Mode
90	57.723661	192.168.1.254	192.168.1.3	ISAKMP	390	Quick Mode
123	73.357116	192.168.1.4	192.168.1.254	ISAKMP	210	Identity Protection (Main Mode)
124	73.358365	192.168.1.254	192.168.1.4	ISAKMP	150	Identity Protection (Main Mode)
125	73.360259	192.168.1.4	192.168.1.254	ISAKMP	414	Identity Protection (Main Mode)
126	73.365109	192.168.1.254	192.168.1.4	ISAKMP	410	Identity Protection (Main Mode)
127	73.370617	192.168.1.4	192.168.1.254	ISAKMP	150	Identity Protection (Main Mode)
128	73.371927	192.168.1.254	192.168.1.4	ISAKMP	118	Identity Protection (Main Mode)
129	73.373833	192.168.1.4	192.168.1.254	ISAKMP	134	Quick Mode
130	73.375309	192.168.1.254	192.168.1.4	ISAKMP	278	Quick Mode
131	73.376987	192.168.1.4	192.168.1.254	ISAKMP	102	Quick Mode
132	73.378729	192.168.1.254	192.168.1.4	ISAKMP	390	Quick Mode

Figure 6: Capture filtered on UDP port 848 after applying the ISAKMP decoder.

5.1.1 IKE Phase 1 – Main Mode

For each Group Member, the capture shows six ISAKMP messages labelled *Identity Protection (Main Mode)* (exchange type 2), constituting the IKE Phase 1 exchange. These six messages serve a single purpose: to establish a mutually authenticated, encrypted channel between the GM and the KS before any key material is exchanged. The three pairs of messages carry out the following steps: the first pair negotiates the SA parameters (encryption algorithm, hash, authentication method, and DH group); the second pair performs the Diffie-Hellman key exchange, from which both sides derive a shared secret; and the third pair carries the encrypted authentication payloads, where each side proves its identity using the pre-shared key `saar`.

A detail visible in the first message of each registration (Figure 7) is that the *Responder SPI* field is `0x0000000000000000`. This is expected: the KS has not yet responded, so no responder cookie has been assigned. By the end of Phase 1, both SPIs are populated, confirming the session is established. Also visible in this first message is the field *Domain of Interpretation: GDOI (2)*, which distinguishes this from a standard IPSec IKE negotiation (DOI 1) and confirms the session is being established specifically to support a GDOI group registration.

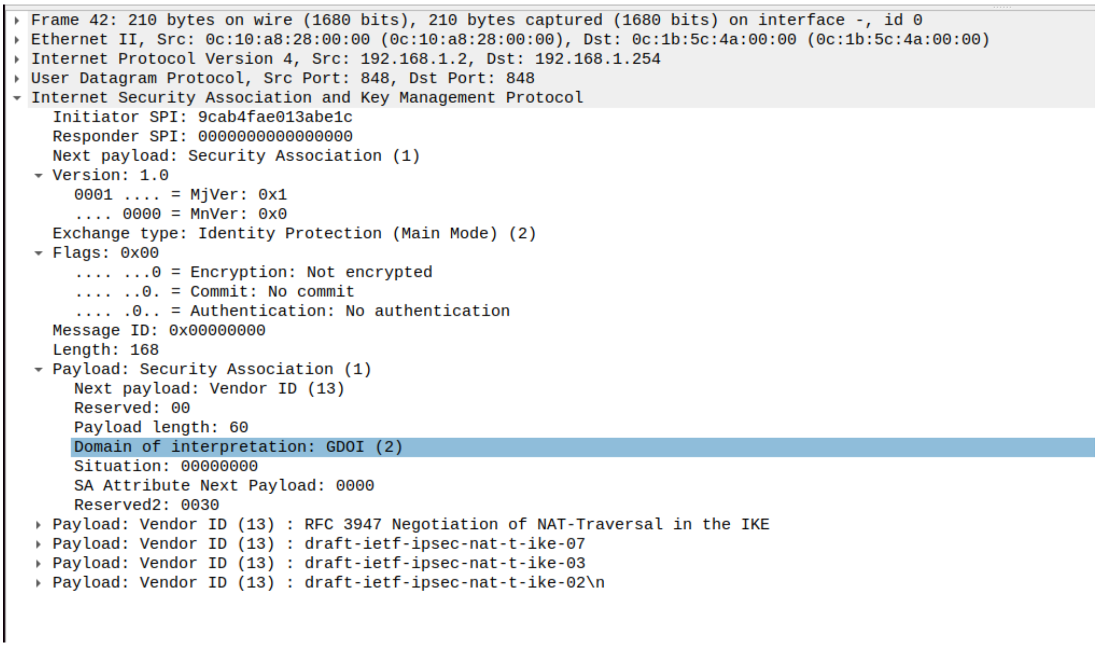


Figure 7: First IKE Phase 1 message expanded. Note the null Responder SPI and `Domain of Interpretation: GDOI (2)`.

5.1.2 GROUPKEY-PULL

Immediately following the six Phase 1 messages, four additional ISAKMP messages appear, labelled by Wireshark as *Quick Mode* (exchange type 32). These four messages constitute the **GROUPKEY-PULL** exchange, through which the GM requests and receives the group keys and IPSec policy from the KS. GDOI reuses the Quick Mode exchange type framing, which is why Wireshark labels them as such; the GDOI nature is confirmed by the DOI field observed in Phase 1.

Figures 8 and 8 show the four packets. All four share the same Message ID (`0x64b4a54f`), confirming they belong to the same exchange. The size of each packet is significant: the GM's initial request (packet 48) carries 64 bytes of encrypted data, while the KS responses carry 208 bytes (packet 49) and 320 bytes (packet 51). This asymmetry reflects the nature of the exchange — the GM sends a short registration request, while the KS response contains the full group policy: the Traffic Encryption Key (TEK), the Key Encryption Key (KEK), the traffic selector ACL, and the rekey parameters. All payloads are encrypted under the Phase 1 SA, so the key material itself is not visible in Wireshark. The fourth message (packet 50, 32 bytes) is the GM's acknowledgement completing the exchange.

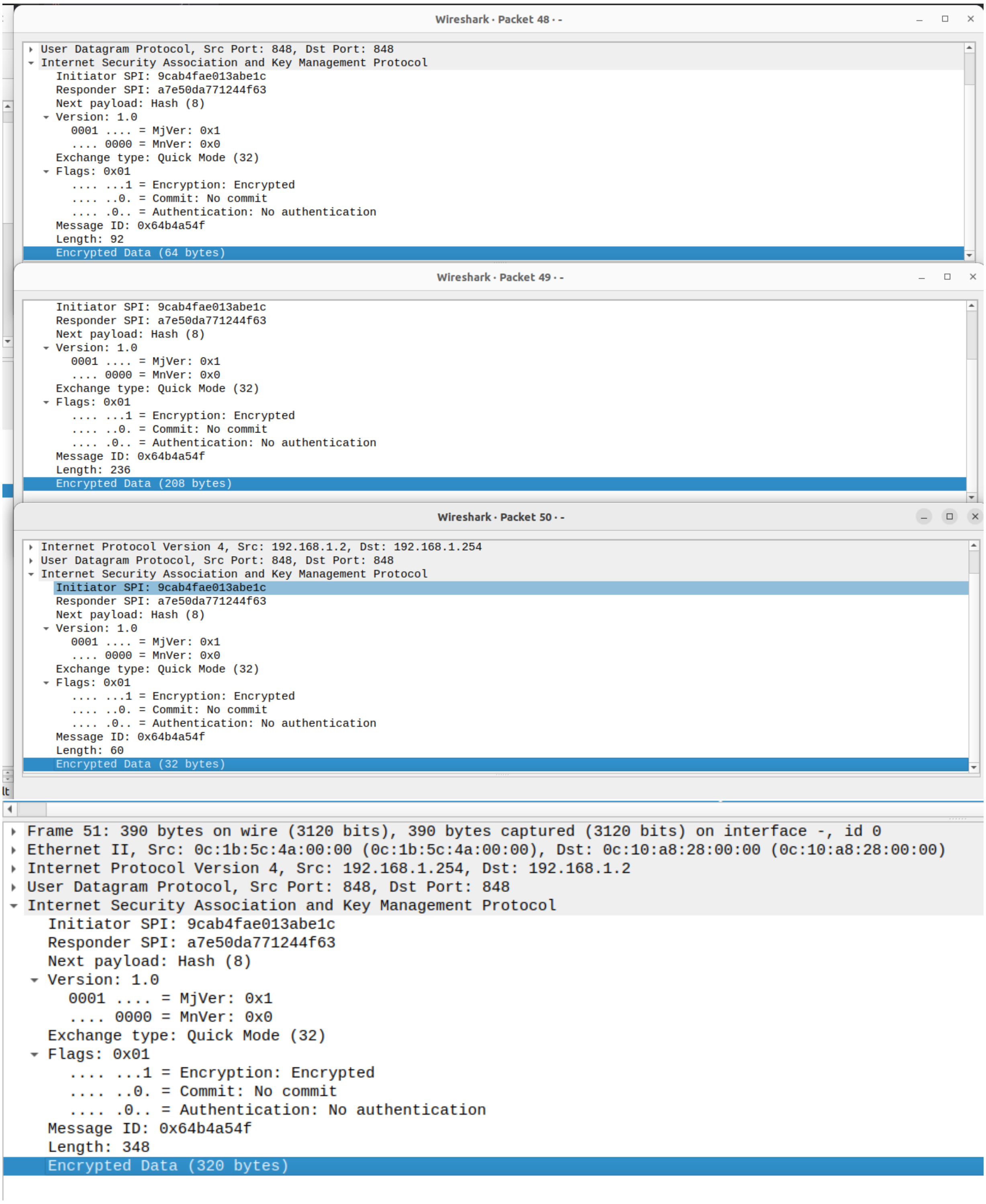


Figure 8: GROUPKEY-PULL exchange (Quick Mode, exchange type 32) between R2 and the Key Server.

5.1.3 Repeated Pattern Across All Group Members

The identical 10-message pattern (6 Main Mode + 4 Quick Mode) was observed independently for each of the three Group Members, as shown in Figures 9, 10, and 11. Each GM performs its own IKE Phase 1 and GROUPKEY-PULL with the KS independently. The KS distributes the same TEK and policy to all three, which is what allows any GM to decrypt traffic encrypted by any other GM.

40	38.588684	192.168.1.254	224.0.0.5	OSPF	94 Hello Packet
41	38.589334	192.168.1.2	192.168.1.254	OSPF	94 Hello Packet
42	40.060597	192.168.1.2	192.168.1.254	ISAKMP	210 Identity Protection (Main Mode)
43	40.062094	192.168.1.254	192.168.1.2	ISAKMP	150 Identity Protection (Main Mode)
44	40.063573	192.168.1.2	192.168.1.254	ISAKMP	414 Identity Protection (Main Mode)
45	40.068368	192.168.1.254	192.168.1.2	ISAKMP	410 Identity Protection (Main Mode)
46	40.073196	192.168.1.2	192.168.1.254	ISAKMP	150 Identity Protection (Main Mode)
47	40.074337	192.168.1.254	192.168.1.2	ISAKMP	118 Identity Protection (Main Mode)
48	40.075796	192.168.1.2	192.168.1.254	ISAKMP	134 Quick Mode
49	40.077498	192.168.1.254	192.168.1.2	ISAKMP	278 Quick Mode
50	40.078929	192.168.1.2	192.168.1.254	ISAKMP	102 Quick Mode
51	40.080272	192.168.1.254	192.168.1.2	ISAKMP	390 Quick Mode
52	40.105590	192.168.1.2	224.0.0.5	OSPF	94 Hello Packet
53	40.963312	192.168.1.254	192.168.1.2	OSPF	78 DB Description
54	45.784618	192.168.1.254	192.168.1.2	OSPF	78 DB Description

Figure 9: 10-message registration sequence for R2 (packets 42–51).

79	56.825291	192.168.1.3	192.168.1.254	OSPF	98 Hello Packet
80	56.825722	192.168.1.254	192.168.1.3	OSPF	78 DB Description
81	57.704422	192.168.1.3	192.168.1.254	ISAKMP	210 Identity Protection (Main Mode)
82	57.705650	192.168.1.254	192.168.1.3	ISAKMP	150 Identity Protection (Main Mode)
83	57.707112	192.168.1.3	192.168.1.254	ISAKMP	414 Identity Protection (Main Mode)
84	57.711980	192.168.1.254	192.168.1.3	ISAKMP	410 Identity Protection (Main Mode)
85	57.717009	192.168.1.3	192.168.1.254	ISAKMP	150 Identity Protection (Main Mode)
86	57.718237	192.168.1.254	192.168.1.3	ISAKMP	118 Identity Protection (Main Mode)
87	57.719704	192.168.1.3	192.168.1.254	ISAKMP	134 Quick Mode
88	57.721103	192.168.1.254	192.168.1.3	ISAKMP	278 Quick Mode
89	57.722382	192.168.1.3	192.168.1.254	ISAKMP	102 Quick Mode
90	57.723661	192.168.1.254	192.168.1.3	ISAKMP	390 Quick Mode
91	57.828611	0c:1b:5c:4a:00:00	0c:1b:5c:4a:00:00	LOOP	60 Reply
92	57.892466	192.168.1.3	224.0.0.5	OSPF	98 Hello Packet
93	59.216737	192.168.1.2	224.0.0.5	OSPF	98 Hello Packet
94	61.708108	192.168.1.254	192.168.1.3	OSPF	78 DB Description
95	63.072595	0c:65:00:07:00:00	Broadcast	ARP	60 Gratuitous ARP for 192.168.1.4 (Reply)

Figure 10: 10-message registration sequence for R3.

121	71.469595	192.168.1.254	192.168.1.4	OSPF	78 DB Description
122	71.672462	0c:65:00:07:00:00	CDP/VTP/DTP/PAgP/UD…	CDP	356 Device ID: R4 Port ID: GigabitEthe
123	73.357116	192.168.1.4	192.168.1.254	ISAKMP	210 Identity Protection (Main Mode)
124	73.358365	192.168.1.254	192.168.1.4	ISAKMP	150 Identity Protection (Main Mode)
125	73.360259	192.168.1.4	192.168.1.254	ISAKMP	414 Identity Protection (Main Mode)
126	73.365109	192.168.1.254	192.168.1.4	ISAKMP	410 Identity Protection (Main Mode)
127	73.370617	192.168.1.4	192.168.1.254	ISAKMP	150 Identity Protection (Main Mode)
128	73.371927	192.168.1.254	192.168.1.4	ISAKMP	118 Identity Protection (Main Mode)
129	73.373833	192.168.1.4	192.168.1.254	ISAKMP	134 Quick Mode
130	73.375309	192.168.1.254	192.168.1.4	ISAKMP	278 Quick Mode
131	73.376987	192.168.1.4	192.168.1.254	ISAKMP	102 Quick Mode
132	73.378729	192.168.1.254	192.168.1.4	ISAKMP	390 Quick Mode
133	74.073061	192.168.1.4	224.0.0.5	OSPF	102 Hello Packet
134	75.710951	192.168.1.254	224.0.0.5	OSPF	102 Hello Packet
135	76.073921	192.168.1.254	192.168.1.4	OSPF	78 DB Description
136	76.606690	192.168.1.3	224.0.0.5	OSPF	102 Hello Packet

Figure 11: 10-message registration sequence for R4.

5.2 SPI Consistency Across Group Members

A fundamental property of GETVPN is that all Group Members share a single Traffic Encryption Key (TEK), meaning all encrypted traffic within the group carries the same Security

Parameter Index (SPI) regardless of source or destination. This was verified by capturing ESP traffic at the link between R2 and the switch while pinging from PC1 to both PC2 and PC3.

5.2.1 Wireshark Capture

Figures 12 and 13 show the ESP packets captured during the two ping sessions. In both cases — PC1 to PC2 (`192.168.10.100` → `192.168.20.100`) and PC1 to PC3 (`192.168.10.100` → `192.168.30.100`) — every ESP packet carries `SPI=0x24FF23B1`. Despite the traffic being destined for different Group Members, the SPI is identical in both captures. Expanding any individual packet confirms `ESP SPI: 0x24ff23b1 (620700593)`.

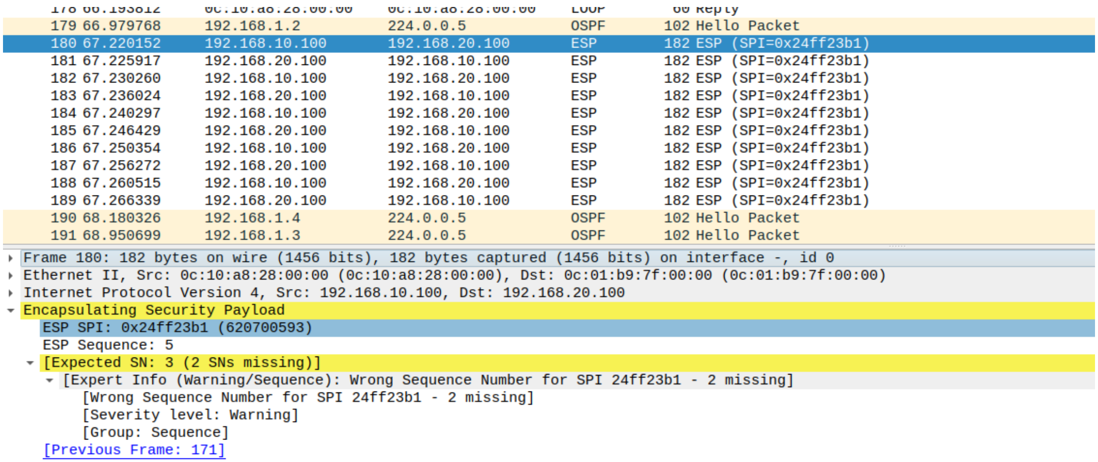


Figure 12: ESP packets captured for PC1 → PC2 ping. All packets carry `SPI=0x24FF23B1`.

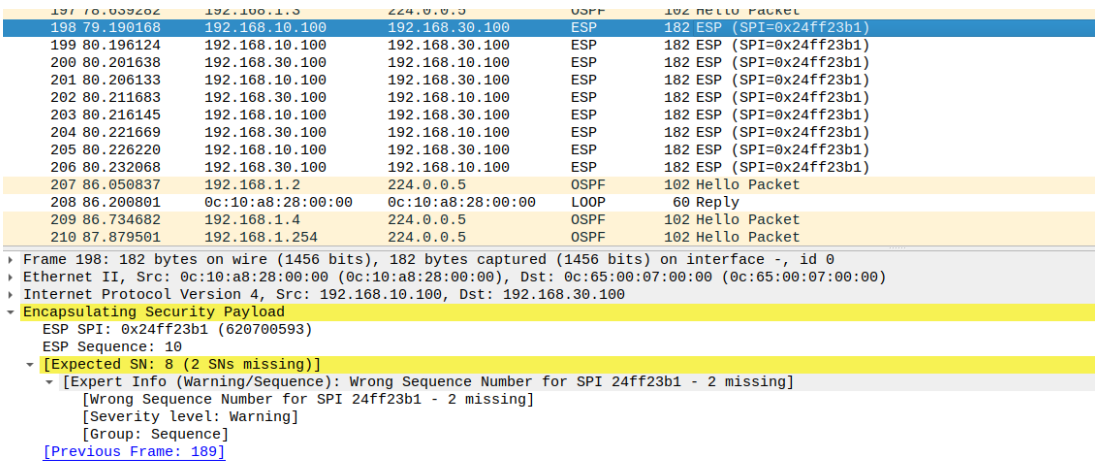
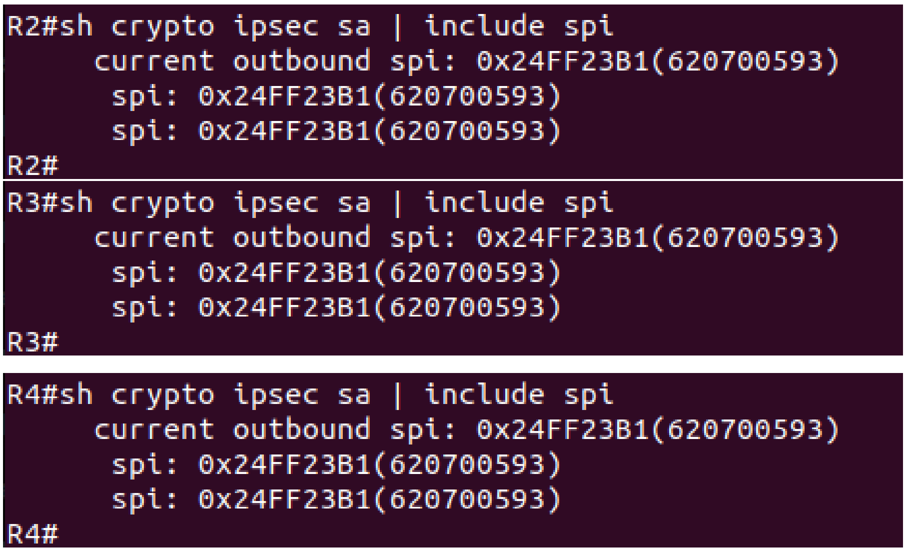


Figure 13: ESP packets captured for PC1 → PC3 ping. Same SPI value despite different destination.

5.2.2 CLI Verification

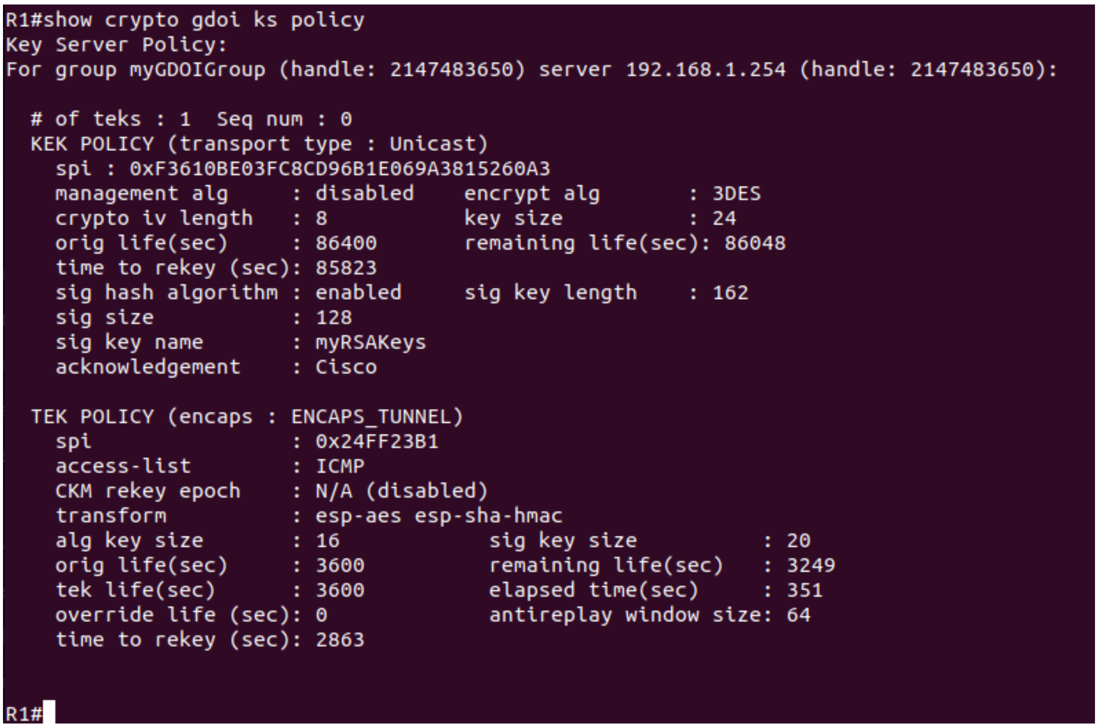
The `show crypto ipsec sa | include spi` command was run on all three Group Members to confirm the shared SPI at the router level:



```
R2#sh crypto ipsec sa | include spi
     current outbound spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
R2#
R3#sh crypto ipsec sa | include spi
     current outbound spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
R3#
R4#sh crypto ipsec sa | include spi
     current outbound spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
R4#
```

Figure 14: `show crypto ipsec sa | include spi` across all Group Members showing that they all have the same SPI

All three GMs report the same SPI value, confirming they hold an identical TEK. The TEK policy at the Key Server, visible in the `show crypto gdoi ks policy` output (Figure 15), confirms that `spi: 0x24FF23B1` is the single TEK distributed to the group, with `transform: esp-aes esp-sha-hmac` and an original lifetime of 3600 seconds.



```
R1#show crypto gdoi ks policy
Key Server Policy:
For group myGDOIGroup (handle: 2147483650) server 192.168.1.254 (handle: 2147483650):

  # of teks : 1  Seq num : 0
  KEK POLICY (transport type : Unicast)
    spi : 0xF3610BE03FC8CD96B1E069A3815260A3
    management alg     : disabled    encrypt alg       : 3DES
    crypto iv length   : 8           key size          : 24
    orig life(sec)     : 86400       remaining life(sec): 86048
    time to rekey (sec): 85823
    sig hash algorithm : enabled     sig key length    : 162
    sig size           : 128
    sig key name       : myRSAKeys
    acknowledgement    : Cisco

  TEK POLICY (encaps : ENCAPS_TUNNEL)
    spi                : 0x24FF23B1
    access-list        : ICMP
    CKM rekey epoch    : N/A (disabled)
    transform          : esp-aes esp-sha-hmac
    alg key size       : 16            sig key size          : 20
    orig life(sec)     : 3600          remaining life(sec)   : 3249
    tek life(sec)      : 3600          elapsed time(sec)     : 351
    override life (sec): 0             antireplay window size: 64
    time to rekey (sec): 2863


R1#
```

Figure 15: `show crypto gdoi ks policy` on R1 confirming the TEK SPI and policy distributed to all GMs.

5.2.3 Why All SPIs Are Identical

In a traditional point-to-point IPSec VPN, each pair of endpoints negotiates its own independent SA with a unique SPI. In GETVPN, the Key Server generates a single TEK and distributes it to all Group Members simultaneously during the GROUPKEY-PULL exchange. Because every GM holds the same key, any GM can decrypt traffic encrypted by any other GM without prior negotiation. The shared SPI is the identifier for this common key, which is why it appears identically on R2, R3, and R4 and in every ESP packet on the wire, regardless of which pair of hosts is communicating.

It is worth noting that the SPI observed here (`0x24FF23B1`) differs from the one seen during the initial verification (`0x6D2F03B9`). This is because the TEK has a lifetime of 3600 seconds and was refreshed between the two observations. The rekey process — in which the KS distributes a new TEK to all GMs before the old one expires — is examined in the following task.

5.3 TEK Rekey Observation

To observe the TEK refresh process, the IPSec SA lifetime on the Key Server was reduced to 900 seconds by adding the following command to the IPSec profile:

```
R1(config)#crypto ipsec profile myIPSecProfile
R1(ipsec-profile)#set security-association lifetime seconds 900
```

Rather than waiting for the timer to expire naturally, the rekey was triggered immediately using the command `crypto gdoi ks rekey` like shown in the Figure 16.

```
R1#crypto gdoi ks rekey
R1#
*Jun  3 15:53:14.253: %GDOI-5-KS_SEND_UNICAST_REKEY: Sending Unicast Rekey with policy-replace for group myGDOIGroup from address 192.168.1.254 with se
q # 1 spi: 0xF3610BE03FC8CD96B1E069A3815260A3
```

Figure 16: Output of the command `crypto gdoi ks rekey` in R1

Figure 17 confirms that the 900 second lifetime was active at the time, with the SA showing a remaining lifetime of 893 seconds and a time to rekey of 777 seconds — well within the configured bound and counting down as expected.

```
      Tagging method           : Disabled
      SA Rekey
         Remaining Lifetime    : 893 secs
        Time to Rekey          : 777 secs
      ACL Configured           : access-list ICMP

     Group Server list         : Local

R1#
```

Figure 17: `show crypto gdoi` on R1 showing the 900 second TEK lifetime in effect after to the manual rekey.

5.3.1 Rekey Messages in Wireshark

Figure 18 shows the Wireshark capture taken at the R1-switch link at the moment the rekey was triggered. Six ISAKMP messages appear in quick succession: three rekey messages from the KS to each GM (packets 47, 49, 51) and three acknowledgements from the GMs back to the KS (packets 48, 50, 52).

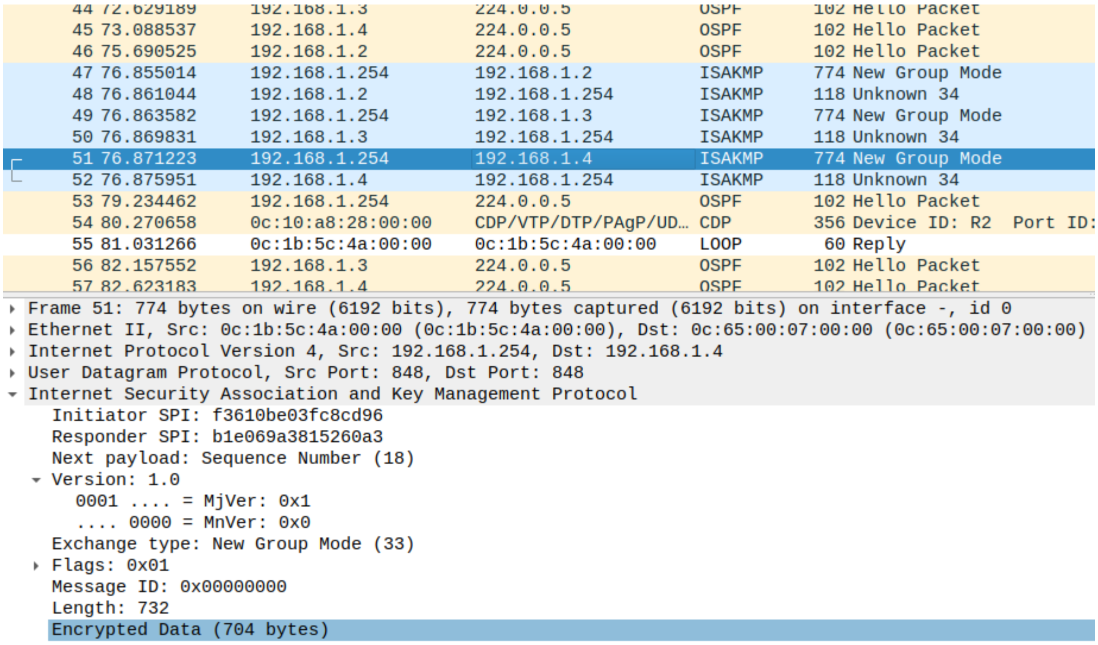


Figure 18: Wireshark capture of the rekey event. Three unicast rekey messages from R1 (`192.168.1.254`) to each GM, followed by acknowledgements.

5.3.2 Exchange Type

Expanding packet 51 (Figure 18) reveals `Exchange type: New Group Mode (33)`. This differs from both the IKE Phase 1 exchange (Main Mode, type 2) and the initial GROUPKEY-PULL (Quick Mode, type 32). New Group Mode (type 33) is the GDOI-specific exchange used exclusively for pushing refreshed key material from a KS to already-registered GMs. The acknowledgement messages use exchange type 34. The direction of the exchange is also reversed compared to GROUPKEY-PULL: here the KS initiates by pushing the new TEK to GMs, rather than GMs pulling from the KS.

All three rekey messages share `Message ID: 0x00000000`, which is the fixed message ID used for GDOI rekey messages. Each packet carries 704 bytes of encrypted data — larger than the original GROUPKEY-PULL responses — as the payload includes the new TEK, updated policy parameters, and the RSA signature used by GMs to verify the authenticity of the rekey message. Since `rekey transport unicast` was configured, the KS sends one independent message to each GM rather than a single multicast.

5.3.3 What Changed

After the rekey completed, the SPI on all GMs changed to a new value, confirming that a new TEK was successfully installed:

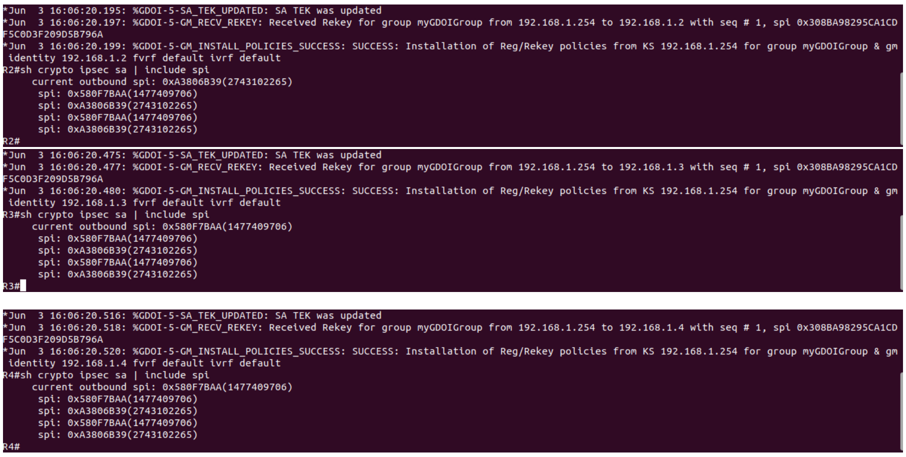


Figure 19: `show crypto ipsec sa | include spi` across all Group Members after the rekeying

The Key Encryption Key (KEK) was not replaced during this event — only the TEK changed. The KEK has a much longer lifetime (86400 seconds) and is used to protect the rekey messages themselves; it is only refreshed on a separate, less frequent schedule. This separation of TEK and KEK lifetimes is a deliberate design choice in GETVPN: TEKs rotate frequently to limit the exposure of any single key, while the KEK rotates less often to reduce management overhead.

6 GETVPN Extensions

6.1 COOP Redundancy

6.1.1 Overview

In the base GETVPN configuration, R1 is the sole Key Server. If R1 becomes unavailable, Group Members can no longer rekey and will eventually lose the ability to encrypt traffic once the current TEK expires. To address this single point of failure, Cisco's **Cooperative Key Server (COOP)** mechanism allows multiple Key Servers to operate together, providing redundancy. The KSs elect a **Primary** and a **Secondary** based on configured priority values. The Primary handles all GM registrations and rekeys under normal operation, while the Secondary maintains synchronised group state via a secure channel between the two KSs. If the Primary becomes unreachable, the Secondary detects the failure through missed heartbeat announcements and promotes itself to Primary, after which GMs re-register with it.

For this extension, a fifth router R5 was added to the topology at address `192.168.1.252`, connected to the same shared segment as the other routers and participating in OSPF. Figure 20 shows the updated topology.

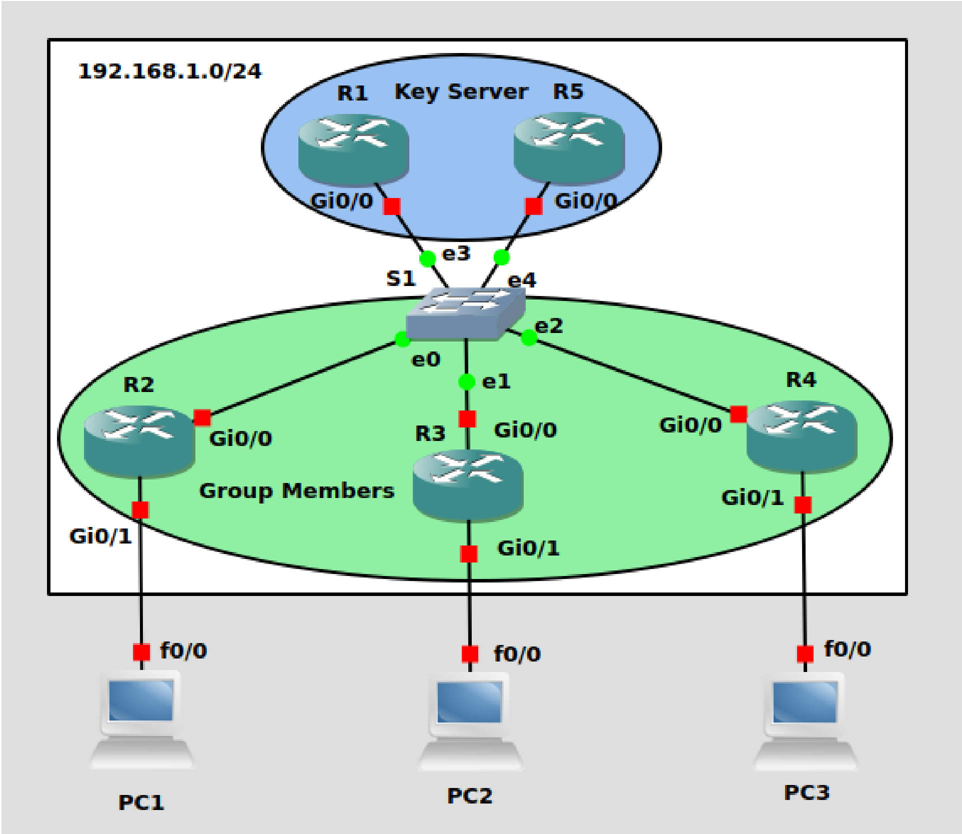


Figure 20: Updated topology with R5 added as the Secondary Key Server.

6.1.2 COOP Configuration

R5 was configured with the same base GETVPN security policy as R1 — identical IKE Phase 1 parameters, transform set, IPSec profile, RSA key pair, traffic ACL, and GDOI group. The COOP-specific addition is the `redundancy` block inside the `server local` configuration, which declares the peer KS address and the local priority:

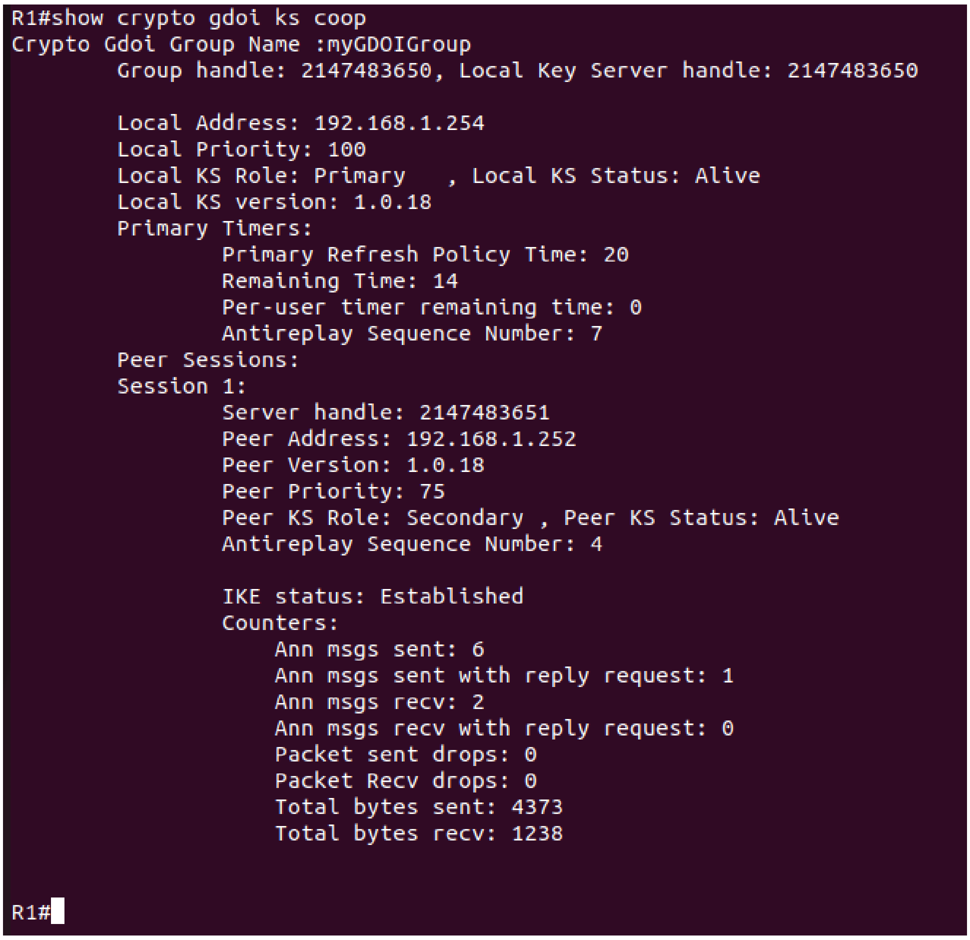
```
! On R1 - add COOP peer and set priority
R1(config)#crypto gdoi group myGDOIGroup
R1(config-gkm-group)#server local
R1(gkm-local-server)#redundancy
R1(gkm-ks-redundancy)#local priority 100
R1(gkm-ks-redundancy)#peer address ipv4 192.168.1.252

! On R5 - same structure, lower priority
R5(config)#crypto gdoi group myGDOIGroup
R5(config-gkm-group)#server local
R5(gkm-local-server)#redundancy
R5(gkm-ks-redundancy)#local priority 75
R5(gkm-ks-redundancy)#peer address ipv4 192.168.1.254
```

R1 was assigned priority 100 and R5 priority 75, making R1 the Primary. Each GM was updated to include R5 as a secondary KS address:

```
Rx(config)#crypto gdoi group myGDOIGroup
Rx(config-gkm-group)#server address ipv4 192.168.1.252
```

6.1.3 Steady-State Verification

Figures 21 and 22 show the COOP status on both KSs under normal operation. R1 reports `Local KS Role: Primary` and R5 reports `Local KS Role: Secondary`, with both peers showing `KS Status: Alive` and `IKE status: Established`. The Primary sends periodic announcement messages to the Secondary every 20 seconds (`Primary Refresh Policy Time: 20`); the Secondary expects these every 30 seconds (`Sec Primary Periodic Time: 30`) and will initiate a role transition if they stop arriving.



```
R1#show crypto gdoi ks coop
Crypto Gdoi Group Name :myGDOIGroup
        Group handle: 2147483650, Local Key Server handle: 2147483650

        Local Address: 192.168.1.254
        Local Priority: 100
        Local KS Role: Primary   , Local KS Status: Alive
        Local KS version: 1.0.18
        Primary Timers:
                Primary Refresh Policy Time: 20
                Remaining Time: 14
                Per-user timer remaining time: 0
                Antireplay Sequence Number: 7
        Peer Sessions:
        Session 1:
                Server handle: 2147483651
                Peer Address: 192.168.1.252
                Peer Version: 1.0.18
                Peer Priority: 75
                Peer KS Role: Secondary , Peer KS Status: Alive
                Antireplay Sequence Number: 4

                IKE status: Established
                Counters:
                    Ann msgs sent: 6
                    Ann msgs sent with reply request: 1
                    Ann msgs recv: 2
                    Ann msgs recv with reply request: 0
                    Packet sent drops: 0
                    Packet Recv drops: 0
                    Total bytes sent: 4373
                    Total bytes recv: 1238


R1#
```

Figure 21: `show crypto gdoi ks coop` on R1. Primary role confirmed, R5 visible as Secondary peer.

```
R5#show crypto gdoi ks coop
Crypto Gdoi Group Name :myGDOIGroup
        Group handle: 2147483650, Local Key Server handle: 2147483650

        Local Address: 192.168.1.252
        Local Priority: 75
        Local KS Role: Secondary , Local KS Status: Alive
        Local KS version: 1.0.18
        Secondary Timers:
                Sec Primary Periodic Time: 30
                Remaining Time: 18, Retries: 0
                Invalid ANN PST recvd: 0
                New GM Temporary Blocking Enforced?: No
                Per-user timer remaining time: 0
                Antireplay Sequence Number: 5
        Peer Sessions:
        Session 1:
                Server handle: 2147483651
                Peer Address: 192.168.1.254
                Peer Version: 1.0.18
                Peer Priority: 100
                Peer KS Role: Primary   , Peer KS Status: Alive
                Antireplay Sequence Number: 6

                IKE status: Established
                Counters:
                    Ann msgs sent: 2
                    Ann msgs sent with reply request: 2
                    Ann msgs recv: 6
                    Ann msgs recv with reply request: 1
                    Packet sent drops: 1
                    Packet Recv drops: 0
                    Total bytes sent: 1350
                    Total bytes recv: 4373


R5#
```

Figure 22: `show crypto gdoi ks coop` on R5. Secondary role confirmed, R1 visible as Primary peer.

The announcement message counters visible in both outputs are complementary — R1 shows 6 sent and 2 received, while R5 shows 2 sent and 6 received — confirming bidirectional synchronisation is working correctly.

Figure 23 shows `show crypto gdoi` on R4 during normal COOP operation. R4 is fully registered with R1 (`Registered with: 192.168.1.254`, `Active TEK Number: 1`) and its `Group Server list` includes both KS addresses — `192.168.1.254` and `192.168.1.252` — confirming that the GM is aware of the Secondary KS and will attempt to contact it if the Primary becomes unavailable.

```
R4#show crypto gdoi
GROUP INFORMATION

    Group Name               : myGDOIGroup
    Group Identity           : 123
    Group Type               : GDOI (ISAKMP)
    Crypto Path              : ipv4
    Key Management Path      : ipv4
    Rekeys received          : 1
    IPSec SA Direction       : Both

     Group Server list       : 192.168.1.254
                               192.168.1.252

Group Member Information For Group myGDOIGroup:
    IPSec SA Direction       : Both
    ACL Received From KS     : gdoi_group_myGDOIGroup_temp_acl

    Group member             : 192.168.1.4     vrf: None
       Local addr/port       : 192.168.1.4/848
       Remote addr/port      : 192.168.1.254/848
       fvrf/ivrf             : None/None
       Version               : 1.0.17
       Registration status   : Registered
       Registered with       : 192.168.1.254
       Re-registers in       : 2216 sec
       Succeeded registration: 2
       Attempted registration: 10
       Last rekey from       : 192.168.1.254
       Last rekey seq num    : 1
       Multicast rekey rcvd  : 0
       DP Error Monitoring   : OFF
       IPSEC init reg executed    : 0
       IPSEC init reg postponed   : 0
       Active TEK Number     : 2
       SA Track (OID/status) : disabled

       allowable rekey cipher: any
       allowable rekey hash  : any
       allowable transformtag: any ESP

    Rekeys cumulative
       Total received        : 1
       After latest register : 0
       Rekey Received        : never

 ACL Downloaded From KS 192.168.1.254:
   access-list   permit icmp any any

TEK POLICY for the current KS-Policy ACEs Downloaded:
  GigabitEthernet0/0:
    IPsec SA:
        spi: 0x73E328C0(1944266944)
        KGS: Disabled
        transform: esp-aes esp-sha-hmac
        sa timing:remaining key lifetime (sec): (2374)
        Anti-Replay(Counter Based) : 64
        tag method : disabled
        alg key size: 16 (bytes)
        sig key size: 20 (bytes)
        encaps: ENCAPS_TUNNEL

    IPsec SA:
        spi: 0xCD59C10D(3445211405)
        KGS: Disabled
        transform: esp-aes esp-sha-hmac
        sa timing:remaining key lifetime (sec): (458)
        Anti-Replay(Counter Based) : 64
        tag method : disabled
        alg key size: 16 (bytes)
        sig key size: 20 (bytes)
        encaps: ENCAPS_TUNNEL

R4#
```

Figure 23: `show crypto gdoi` on R4 under normal COOP operation. Registered with R1 as Primary, both KS addresses present in the server list, TEK active.

6.2 Failover Test – First Attempt and Diagnosis

To test the failover behaviour, R1's `gi0/0` interface was shut down while Wireshark probes captured traffic on both the R1-switch and R5-switch links. The KS-level transition worked immediately and correctly: within 30 seconds of R1 becoming unreachable, R5 produced the console messages shown in Figure 24, confirming it detected the failure and promoted itself to Primary.

```
R5#
*Jun  4 16:05:42.710: %GDOI-3-COOP_KS_UNREACH: Cooperative KS 192.168.1.254 Unreachable in group myGDOIGroup. IKE SA Status = Failed to establish.
R5#
*Jun  4 16:05:42.710: %GDOI-5-COOP_KS_TRANS_TO_PRI: KS 192.168.1.252 in group myGDOIGroup transitioned to Primary (Previous Primary = 192.168.1.254)
```

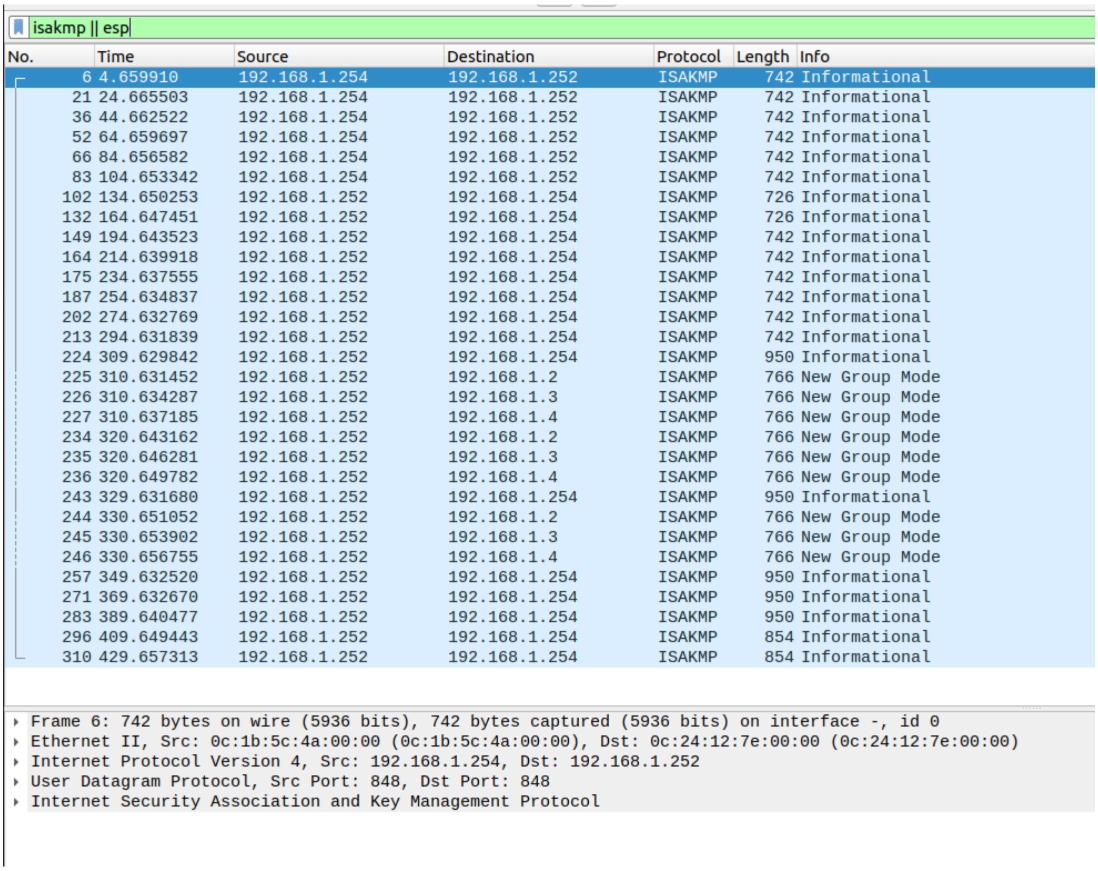
Figure 24: R5 console at the moment of failover. R1 is declared unreachable and R5 transitions to Primary.

Figure 25 confirms the COOP status on R5 after the transition: `Local KS Role: Primary` and `Peer KS Status: Dead`.

```
R5#show crypto gdoi ks coop
Crypto Gdoi Group Name :myGDOIGroup
        Group handle: 2147483650, Local Key Server handle: 2147483650

        Local Address: 192.168.1.252
        Local Priority: 75
        Local KS Role: Primary   , Local KS Status: Alive
        Local KS version: 1.0.18
        Primary Timers:
                Primary Refresh Policy Time: 20
                Remaining Time: 6
                Per-user timer remaining time: 0
                Antireplay Sequence Number: 9
        Peer Sessions:
        Session 1:
                Server handle: 2147483651
                Peer Address: 192.168.1.254
                Peer Version: 1.0.18
                Peer Priority: Unknown
                Peer KS Role: Primary   , Peer KS Status: Dead
                Antireplay Sequence Number: 62

                IKE status: Established
                Counters:
                    Ann msgs sent: 2
                    Ann msgs sent with reply request: 5
                    Ann msgs recv: 62
                    Ann msgs recv with reply request: 1
                    Packet sent drops: 1
                    Packet Recv drops: 0
                    Total bytes sent: 3215
                    Total bytes recv: 39980
```

Figure 25: `show crypto gdoi ks coop` on R5 after R1 failure. R5 has assumed the Primary role.

Figure 26 shows the R5-switch capture. The early Informational packets are the COOP heartbeats between R1 and R5 during normal operation. After the heartbeats from R1 stop, R5 sends New Group Mode rekey messages (exchange type 33) to all three GMs, announcing itself as the new active Key Server.



isakmp || esp

No.	Time	Source	Destination	Protocol	Length	Info
6	4.659910	192.168.1.254	192.168.1.252	ISAKMP	742	Informational
21	24.665503	192.168.1.254	192.168.1.252	ISAKMP	742	Informational
36	44.662522	192.168.1.254	192.168.1.252	ISAKMP	742	Informational
52	64.659697	192.168.1.254	192.168.1.252	ISAKMP	742	Informational
66	84.656582	192.168.1.254	192.168.1.252	ISAKMP	742	Informational
83	104.653342	192.168.1.254	192.168.1.252	ISAKMP	742	Informational
102	134.650253	192.168.1.252	192.168.1.254	ISAKMP	726	Informational
132	164.647451	192.168.1.252	192.168.1.254	ISAKMP	726	Informational
149	194.643523	192.168.1.252	192.168.1.254	ISAKMP	742	Informational
164	214.639918	192.168.1.252	192.168.1.254	ISAKMP	742	Informational
175	234.637555	192.168.1.252	192.168.1.254	ISAKMP	742	Informational
187	254.634837	192.168.1.252	192.168.1.254	ISAKMP	742	Informational
202	274.632769	192.168.1.252	192.168.1.254	ISAKMP	742	Informational
213	294.631839	192.168.1.252	192.168.1.254	ISAKMP	742	Informational
224	309.629842	192.168.1.252	192.168.1.254	ISAKMP	950	Informational
225	310.631452	192.168.1.252	192.168.1.2	ISAKMP	766	New Group Mode
226	310.634287	192.168.1.252	192.168.1.3	ISAKMP	766	New Group Mode
227	310.637185	192.168.1.252	192.168.1.4	ISAKMP	766	New Group Mode
234	320.643162	192.168.1.252	192.168.1.2	ISAKMP	766	New Group Mode
235	320.646281	192.168.1.252	192.168.1.3	ISAKMP	766	New Group Mode
236	320.649782	192.168.1.252	192.168.1.4	ISAKMP	766	New Group Mode
243	329.631680	192.168.1.252	192.168.1.254	ISAKMP	950	Informational
244	330.651052	192.168.1.252	192.168.1.2	ISAKMP	766	New Group Mode
245	330.653902	192.168.1.252	192.168.1.3	ISAKMP	766	New Group Mode
246	330.656755	192.168.1.252	192.168.1.4	ISAKMP	766	New Group Mode
257	349.632520	192.168.1.252	192.168.1.254	ISAKMP	950	Informational
271	369.632670	192.168.1.252	192.168.1.254	ISAKMP	950	Informational
283	389.640477	192.168.1.252	192.168.1.254	ISAKMP	950	Informational
296	409.649443	192.168.1.252	192.168.1.254	ISAKMP	854	Informational
310	429.657313	192.168.1.252	192.168.1.254	ISAKMP	854	Informational

```
Frame 6: 742 bytes on wire (5936 bits), 742 bytes captured (5936 bits) on interface -, id 0
Ethernet II, Src: 0c:1b:5c:4a:00:00 (0c:1b:5c:4a:00:00), Dst: 0c:24:12:7e:00:00 (0c:24:12:7e:00:00)
Internet Protocol Version 4, Src: 192.168.1.254, Dst: 192.168.1.252
User Datagram Protocol, Src Port: 848, Dst Port: 848
Internet Security Association and Key Management Protocol
```

Figure 26: R5-switch Wireshark capture showing COOP heartbeats, R5 promotion, and New Group Mode rekeys to GMs.

Despite the clean KS-level transition, the Group Members failed to complete re-registration with R5. Figure 27 shows R5's member table still reflecting the state synchronised from R1, with all GMs listed as registered but with `Key Server ID: 192.168.1.254` — no fresh registration with R5 had completed.

```
R5#show crypto gdoi ks members

Group Member Information :

Number of rekeys sent for group myGDOIGroup : 0
Number of retransmits during the last rekey for group myGDOIGroup : 0
Duration of the last rekey for group myGDOIGroup : 0 msec


Group Member ID    : 192.168.1.2         GM Version: 1.0.17
 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0

Group Member ID    : 192.168.1.3         GM Version: 1.0.17
```

```
Group Member Information :

 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0

Group Member ID    : 192.168.1.4         GM Version: 1.0.17
 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

Group Member Information :


 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0


R5#
```

Figure 27: `show crypto gdoi ks members` on R5 after first failover attempt. GMs appear registered from synchronised state but no fresh registration with R5 has completed.

6.2.1 Root Cause Diagnosis

To diagnose the failure, `debug crypto isakmp` was enabled on R2 and `clear crypto gdoi` was used to force an immediate re-registration attempt. The debug output revealed two distinct problems occurring in sequence.

The first problem was that R2's re-registration timer had backed off to nearly 50 minutes after 10 failed attempts, so it was not actively retrying. When forced, R2 first attempted to reach R1 at `192.168.1.254`, which naturally failed since R1 was down (Figure 28). After exhausting retransmits it fell through to trying R5 at `192.168.1.252`, where the second and critical problem appeared (Figure 29):

```
R2#
*Jun  6 10:36:08.275: ISAKMP: (0):retransmitting phase 1 MM_NO_STATE...
*Jun  6 10:36:08.275: ISAKMP: (0):: incrementing error counter on sa, attempt 1 of 3: retransmit phase 1
*Jun  6 10:36:08.275: ISAKMP: (0):retransmitting phase 1 MM_NO_STATE
*Jun  6 10:36:08.275: ISAKMP-PAK: (0):sending packet to 192.168.1.254 my_port 848 peer_port 848 (I) MM_NO_STATE
*Jun  6 10:36:08.275: ISAKMP: (0):Sending an IKE IPv4 Packet.
R2#
*Jun  6 10:36:18.275: ISAKMP: (0):retransmitting phase 1 MM_NO_STATE...
*Jun  6 10:36:18.275: ISAKMP: (0):: incrementing error counter on sa, attempt 2 of 3: retransmit phase 1
*Jun  6 10:36:18.275: ISAKMP: (0):retransmitting phase 1 MM_NO_STATE
*Jun  6 10:36:18.275: ISAKMP-PAK: (0):sending packet to 192.168.1.254 my_port 848 peer_port 848 (I) MM_NO_STATE
*Jun  6 10:36:18.275: ISAKMP: (0):Sending an IKE IPv4 Packet.
R2#
*Jun  6 10:36:28.275: ISAKMP: (0):retransmitting phase 1 MM_NO_STATE...
*Jun  6 10:36:28.275: ISAKMP: (0):: incrementing error counter on sa, attempt 3 of 3: retransmit phase 1
*Jun  6 10:36:28.275: ISAKMP: (0):retransmitting phase 1 MM_NO_STATE
*Jun  6 10:36:28.275: ISAKMP-PAK: (0):sending packet to 192.168.1.254 my_port 848 peer_port 848 (I) MM_NO_STATE
*Jun  6 10:36:28.275: ISAKMP: (0):Sending an IKE IPv4 Packet.
R2#
```

Figure 28: R2 debug output showing IKE Phase 1 retransmission attempts to R1 (`192.168.1.254`) failing with no response.

```
*Jun  6 10:36:38.277: ISAKMP: (0):Can not start Aggressive mode, trying Main mode.
*Jun  6 10:36:38.277: ISAKMP-ERROR: (0):No pre-shared key with 192.168.1.252!
*Jun  6 10:36:38.277: ISAKMP-ERROR: (0):No Cert or pre-shared address key.
*Jun  6 10:36:38.277: ISAKMP-ERROR: (0):cons
R2#truct_initial_message: Can not start Main mode
```

Figure 29: R2 debug output showing the explicit error when attempting IKE Phase 1 with R5: `No pre-shared key with 192.168.1.252`.

The error `No pre-shared key with 192.168.1.252` identifies the root cause precisely: the GMs had a pre-shared key configured only for R1's address (`192.168.1.254`), not for R5's address (`192.168.1.252`). Without a matching credential, IKE Phase 1 cannot start and the GROUPKEY-PULL exchange cannot take place. R5 was correctly promoted to Primary and correctly sending rekey messages, but the GMs had no way to authenticate with it.

The fix was straightforward — add the pre-shared key for R5's address on each GM:

```
R2(config)#crypto isakmp key saar address 192.168.1.252
R3(config)#crypto isakmp key saar address 192.168.1.252
R4(config)#crypto isakmp key saar address 192.168.1.252
```

This was a configuration omission rather than a fundamental limitation: the same pre-shared key `saar` is used, simply extended to cover R5's address in addition to R1's. In the original base configuration, GMs only needed to authenticate with one KS, so only one address was configured. COOP requires GMs to be able to authenticate with any KS in the group.

6.3 Clean Failover Verification

With the pre-shared key added on all GMs, the failover test was repeated. R1's interface was shut down again. R5 promoted itself to Primary as before. This time, when the GMs attempted re-registration with R5, IKE Phase 1 completed successfully and the full GROUPKEY-PULL exchange took place.

Figures 30 and 30 show R2's `show crypto gdoi` output after successful re-registration with R5, confirming `Registration status: Registered`, `Registered with: 192.168.1.252`, and `Active TEK Number: 1`.

```
R2#sh crypto gdoi
GROUP INFORMATION

    Group Name               : myGDOIGroup
    Group Identity           : 123
    Group Type               : GDOI (ISAKMP)
    Crypto Path              : ipv4
    Key Management Path      : ipv4
    Rekeys received          : 0
    IPSec SA Direction       : Both

     Group Server list       : 192.168.1.254
                               192.168.1.252

Group Member Information For Group myGDOIGroup:
    IPSec SA Direction       : Both
    ACL Received From KS     : gdoi_group_myGDOIGroup_temp_acl

    Group member             : 192.168.1.2      vrf: None
       Local addr/port       : 192.168.1.2/848
       Remote addr/port      : 192.168.1.252/848
       fvrf/ivrf             : None/None
       Version               : 1.0.17
       Registration status   : Registered
       Registered with       : 192.168.1.252
       Re-registers in       : 2441 sec
       Succeeded registration: 1
       Attempted registration: 2
       Last rekey from       : 0.0.0.0
       Last rekey seq num    : 156
       Unicast rekey received: 0
       Rekey ACKs sent       : 0
       Rekey Received        : never
       DP Error Monitoring   : OFF
       IPSEC init reg executed    : 0
       IPSEC init reg postponed   : 0
       Active TEK Number     : 1
       SA Track (OID/status) : disabled

       allowable rekey cipher: any
       allowable rekey hash  : any
       allowable transformtag: any ESP

    Rekeys cumulative
       Total received        : 0
       After latest register : 0
       Rekey Acks sents      : 0

KEK POLICY:
    Rekey Transport Type     : Unicast
    Lifetime (secs)          : 84638
    Encrypt Algorithm        : 3DES
    Key Size                 : 192
    Sig Hash Algorithm       : HMAC_AUTH_SHA
    Sig Key Length (bits)    : 1296

TEK POLICY for the current KS-Policy ACEs Downloaded:
  GigabitEthernet0/0:
    IPsec SA:
        spi: 0x2094101E(546574366)
        KGS: Disabled
        transform: esp-aes esp-sha-hmac
        sa timing:remaining key lifetime (sec): (2626)
        Anti-Replay(Counter Based) : 64
        tag method : disabled
        alg key size: 16 (bytes)
        sig key size: 20 (bytes)
        encaps: ENCAPS_TUNNEL

    IPsec SA:
        spi: 0x9C0EC759(2618214233)
        KGS: Disabled
        transform: esp-aes esp-sha-hmac
        sa timing:remaining key lifetime (sec): expired
        Anti-Replay(Counter Based) : 64
        tag method : disabled
        alg key size: 16 (bytes)
        sig key size: 20 (bytes)
        encaps: ENCAPS_TUNNEL

R2#
```

Figure 30: `show crypto gdoi` on R2 after successful failover. Registered with R5 (`192.168.1.252`), TEK active.

Figure 31 shows R5's member table with all three GMs now showing fresh registrations with `Key Server ID: 192.168.1.252`.

```
R5#sh crypto gdoi ks members

Group Member Information :

Number of rekeys sent for group myGDOIGroup : 2
Number of retransmits during the last rekey for group myGDOIGroup : 0
Duration of the last rekey for group myGDOIGroup : 13 msec

Group Member ID    : 192.168.1.2        GM Version: 1.0.17
 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.252
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0

Group Member ID    : 192.168.1.3        GM Version: 1.0.17
 Group ID          : 123

Group Member Information :

 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.252
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0
```

```
 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0

Group Member ID    : 192.168.1.4        GM Version: 1.0.17
 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.252
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0


Group Member Information :

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0


R5#
```

Figure 31: `show crypto gdoi ks members` on R5 after successful failover. All three GMs registered with R5 as the active Key Server.

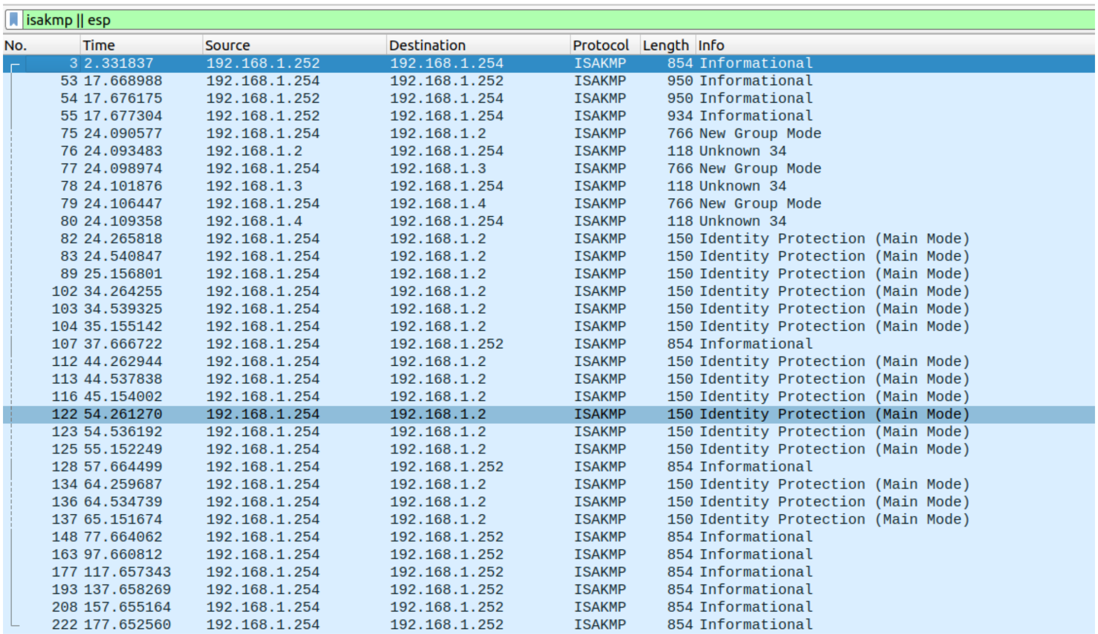
Traffic encryption was verified by pinging from PC1 to PC2 and PC3 with R5 as the active Primary KS, shown in Figure 32. Both pings succeeded, confirming that the GMs received a valid TEK from R5 and are encrypting traffic correctly.

```
PC1#ping 192.168.20.100

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.100, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 8/12/20 ms
PC1#
```

Figure 32: Successful pings from PC1 to PC2 and PC3 with R5 as Primary KS, confirming traffic encryption is operational after failover.

6.4 Recovery – R1 Returning as Primary

When R1's interface was brought back up, it re-established the COOP channel with R5 and reclaimed the Primary role due to its higher priority (100 vs 75). Figure 33 shows the R1-switch capture at this moment: R1 and R5 exchange Informational COOP announcements to re-establish the channel, R1 then immediately sends New Group Mode rekey messages to all three GMs (packets 75, 77, 79) announcing its return as Primary, and the GMs respond with acknowledgements (Unknown 34). This is followed by fresh Identity Protection (Main Mode) exchanges as the GMs re-register with R1.

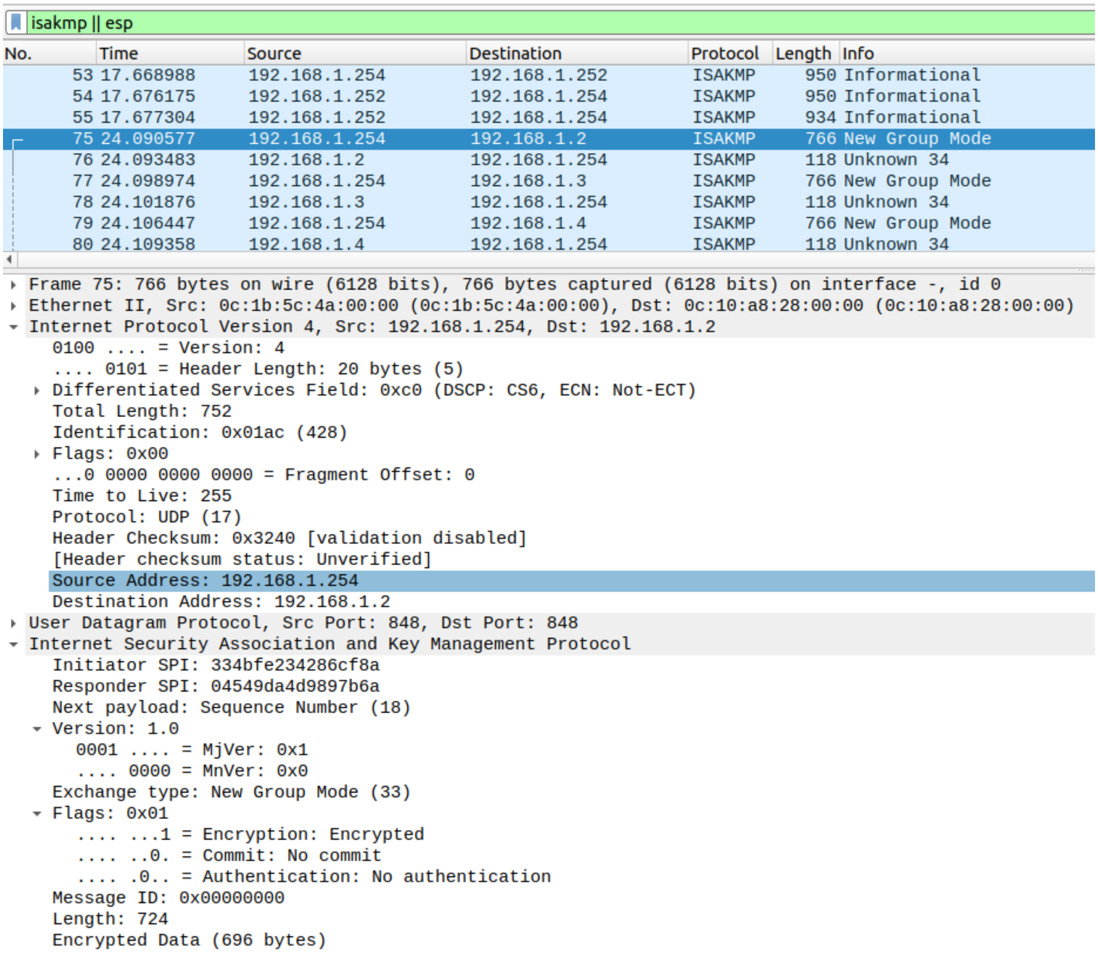


isakmp || esp

No.	Time	Source	Destination	Protocol	Length	Info
3	2.331837	192.168.1.252	192.168.1.254	ISAKMP	854	Informational
53	17.668988	192.168.1.254	192.168.1.252	ISAKMP	950	Informational
54	17.676175	192.168.1.252	192.168.1.254	ISAKMP	950	Informational
55	17.677304	192.168.1.252	192.168.1.254	ISAKMP	934	Informational
75	24.090577	192.168.1.254	192.168.1.2	ISAKMP	766	New Group Mode
76	24.093483	192.168.1.2	192.168.1.254	ISAKMP	118	Unknown 34
77	24.098974	192.168.1.254	192.168.1.3	ISAKMP	766	New Group Mode
78	24.101876	192.168.1.3	192.168.1.254	ISAKMP	118	Unknown 34
79	24.106447	192.168.1.254	192.168.1.4	ISAKMP	766	New Group Mode
80	24.109358	192.168.1.4	192.168.1.254	ISAKMP	118	Unknown 34
82	24.265818	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
83	24.540847	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
89	25.156801	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
102	34.264255	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
103	34.539325	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
104	35.155142	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
107	37.666722	192.168.1.254	192.168.1.252	ISAKMP	854	Informational
112	44.262944	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
113	44.537838	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
116	45.154002	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
122	54.261270	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
123	54.536192	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
125	55.152249	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
128	57.664499	192.168.1.254	192.168.1.252	ISAKMP	854	Informational
134	64.259687	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
136	64.534739	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
137	65.151674	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
148	77.664062	192.168.1.254	192.168.1.252	ISAKMP	854	Informational
163	97.660812	192.168.1.254	192.168.1.252	ISAKMP	854	Informational
177	117.657343	192.168.1.254	192.168.1.252	ISAKMP	854	Informational
193	137.658269	192.168.1.254	192.168.1.252	ISAKMP	854	Informational
208	157.655164	192.168.1.254	192.168.1.252	ISAKMP	854	Informational
222	177.652560	192.168.1.254	192.168.1.252	ISAKMP	854	Informational

Figure 33: R1-switch capture showing the recovery sequence: COOP re-establishment, R1 sending New Group Mode rekeys to GMs, and GMs re-registering with R1 as Primary.

Figure 34 shows an expanded New Group Mode packet from R1, confirming exchange type 33 and the direction R1 → GM.



isakmp || esp

No.	Time	Source	Destination	Protocol	Length	Info
53	17.668988	192.168.1.254	192.168.1.252	ISAKMP	950	Informational
54	17.676175	192.168.1.252	192.168.1.254	ISAKMP	950	Informational
55	17.677304	192.168.1.252	192.168.1.254	ISAKMP	934	Informational
75	24.090577	192.168.1.254	192.168.1.2	ISAKMP	766	New Group Mode
76	24.093483	192.168.1.2	192.168.1.254	ISAKMP	118	Unknown 34
77	24.098974	192.168.1.254	192.168.1.3	ISAKMP	766	New Group Mode
78	24.101876	192.168.1.3	192.168.1.254	ISAKMP	118	Unknown 34
79	24.106447	192.168.1.254	192.168.1.4	ISAKMP	766	New Group Mode
80	24.109358	192.168.1.4	192.168.1.254	ISAKMP	118	Unknown 34

```
Frame 75: 766 bytes on wire (6128 bits), 766 bytes captured (6128 bits) on interface -, id 0
Ethernet II, Src: 0c:1b:5c:4a:00:00 (0c:1b:5c:4a:00:00), Dst: 0c:10:a8:28:00:00 (0c:10:a8:28:00:00)
Internet Protocol Version 4, Src: 192.168.1.254, Dst: 192.168.1.2
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0xc0 (DSCP: CS6, ECN: Not-ECT)
  Total Length: 752
  Identification: 0x01ac (428)
  Flags: 0x00
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 255
  Protocol: UDP (17)
  Header Checksum: 0x3240 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.254
  Destination Address: 192.168.1.2
User Datagram Protocol, Src Port: 848, Dst Port: 848
Internet Security Association and Key Management Protocol
  Initiator SPI: 334bfe234286cf8a
  Responder SPI: 04549da4d9897b6a
  Next payload: Sequence Number (18)
  Version: 1.0
    0001 .... = MjVer: 0x1
    .... 0000 = MnVer: 0x0
  Exchange type: New Group Mode (33)
  Flags: 0x01
    .... ...1 = Encryption: Encrypted
    .... ..0. = Commit: No commit
    .... .0.. = Authentication: No authentication
  Message ID: 0x00000000
  Length: 724
  Encrypted Data (696 bytes)
```

Figure 34: Expanded New Group Mode packet (exchange type 33) from R1 to a GM during the recovery sequence.

7 Conclusion